Python(22AIE205) Programming Lab

Weekly Lab Report-4

LAB 4: Patterns and Lists in python

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Department:	Artificial Intelligence and Engineering
Experiment Date:	28-07-2026

Experiment Record:

Date:	28-07-2026
Experiment Name:	Patterns and Lists in python
Aim:	To practice the printing of different patterns and Lists in python using Python IDLE.
Software Required:	Python 3.x and Python IDLE

Questions Given:

Q1. Write a python program to print the following patterns

a)
```
1
1 2
1 2 3
1 2 3 4
```

b)
```
A
A B
A B C
A B C D
```

c)
```
      1
    1 2
  1 2 3
```

d)
```
      *
    * * *
  * * * * *
    * * *
      *
```

Q2. Write a Python program to shift each character of a given string by one position in the ASCII sequence.

Q3. Write a Python program to read a specified number of elements from the user through the keyboard and store them in a list. The program should: 1. Prompt the user to enter the number of elements, n. 2. Use a loop to accept n elements one by one from the user. 3. Append each entered element to a list. 4. Display the final list after all elements have been entered.

Q4. Write a Python program to read temperature values for each day of the week (Monday to Sunday) from the user and store them in a list. Find the maximum and minimum temperature recorded during the week, along with the day on which it occurred.

Q5. Write a Python program to remove duplicate elements from a list by preserving the order.

Q6. Create a program that manages a grocery shopping list. The program should:

- Initialize a list with some grocery items.
- Add a new item to the list.
- Remove an item from the list.
- Print the final shopping list.

Q7. Write a menu-driven Python program to manage a list of books in a library using built-in list methods (no user-defined functions). The program

should display a menu repeatedly (using a loop) and perform the operation selected by the user, until the user chooses to exit.

Menu:

Library Management System

1. Display all books
2. Add a new book
3. Insert a book at a specific position
4. Check availability of a book
5. Count occurrences of a book
6. Remove a book
7. Sort books alphabetically
8. Exit

Enter your choice:

Requirements:

Use a while True loop to keep showing the menu until the user selects Exit.

Use input() to take the user's choice and any required book title/position.

Use the appropriate list method for each operation (append(), insert(), index(), count(), remove(), sort()).

Handle invalid choices and cases where a book isn't found (e.g., using if title in books before calling index() or remove()).

Use enumerate() to display books with numbering wherever the list is shown to the user — specifically:

When displaying all books (option 1)

When showing the current list before asking for an insert position (option 3), so the user can pick the correct index

When displaying the sorted list (option 7)

Q8. Write a Python program to demonstrate the use of list comprehension for the following tasks:

a) Given a list of numbers, create a new list containing only the even numbers.

b) Given a list of numbers, create a new list containing the square of each number.

c) Given a list of strings (book titles), create a new list containing only the titles that have more than 10 characters.

d) Given a list of numbers, create a new list containing the square of only the even numbers (combine filtering and transformation in one comprehension).

Programs, Sample Outputs and Results:

Question 1: Patterns

Program

pattern1.py - C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\Lab4\pattern1.py (3.14.6)

File Edit Format Run Options Window Help

```
for i in range(1, 5):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

pattern2.py - C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\Lab4\pattern2.py (3.14.6)

File Edit Format Run Options Window Help

```
for i in range(1, 5):
    for j in range(i):
        print(chr(65 + j), end=" ")
    print()
```

pattern3.py - C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\Lab4\pattern3....

File Edit Format Run Options Window Help

```
for i in range(1, 4):
    print(" " * (4 - i), end="")
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

pattern4.py - C:/Users/prana/OneDrive/Desktop/AV.SC.U4AIE25102/Lab4/pattern4.py (3.14.6)

File Edit Format Run Options Window Help

```
for i in range(3):
    print("  " * (2 - i), end="")
    print("* " * (2 * i + 1))

for i in range(1, 3):
    print("  " * i, end="")
    print("* " * (5 - 2 * i))
```

Input and Output:

```
== RESTART: C:/Users/prana/OneDrive/Desktop/AV.SC.U4AIE25102/Lab4/pattern1.py ==
1
1 2
1 2 3
1 2 3 4

== RESTART: C:/Users/prana/OneDrive/Desktop/AV.SC.U4AIE25102/Lab4/pattern2.py ==
A
A B
A B C
A B C D

== RESTART: C:/Users/prana/OneDrive/Desktop/AV.SC.U4AIE25102/Lab4/pattern3.py ==
    1
  1 2
1 2 3

== RESTART: C:/Users/prana/OneDrive/Desktop/AV.SC.U4AIE25102/Lab4/pattern4.py ==
  *
 * * *
* * * * *
 * * *
  *

== RESTART: C:/Users/prana/OneDrive/Desktop/AV.SC.U4AIE25102/Lab4/pattern4.py ==
    *
  * * *
* * * * *
  * * *
    *
```

Result: The program executed successfully by displaying the patterns

Question-2: . Write a Python program to shift each character of a given string by one position in the ASCII sequence.

Python Code:

```
text=input("Enter the Text:")
shifted =""
for ch in text:
    shifted += chr((ord(ch)+1))
print("Shifted string:",shifted)
```

Input&Output:

```
==== RESTART: C:\Users\prana\OneDrive\Desktop
Enter the Text:hlo
Shifted string: imp
```

Result:The program is executed successfully ,by taking input from and shift each character of a given string by one position in the ASCII sequence.

Question-3: Write a Python program to read a specified number of elements from the user through the keyboard and store them in a list. The program should: 1. Prompt the user to enter the number of elements, n. 2. Use a loop to accept n elements one by one from the user. 3. Append each entered element to a list. 4. Display the final list after all elements have been entered.

Program:

```
n=int(input("Enter the Number of elements to add:"))
lst=[]
for i in range(n):
    element=input("Enter element:")
    lst.append(element)
    print("The list is:",lst)
```

Input & Output:

```
= RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\Lab4
\listadd.py
Enter the Number of elements to add:3
Enter element:pranav
The list is: ['pranav']
Enter element:good
The list is: ['pranav', 'good']
Enter element:boy
The list is: ['pranav', 'good', 'boy']
```

Result:The program is executed successfully ,by taking no of elements from user and elements to append each.

Question-4 Write a Python program to read temperature values for each day of the week (Monday to Sunday) from the user and store them in a list. Find the maximum and minimum temperature recorded during the week, along with the day on which it occurred

Program ,Input&Output:

:

```
days=["monday","tuesday","wednesday","thursday","friday","saturday","sunday"]
temp=[]
for i in range(7):
    t=float(input("Enter the temperature of " + days[i]  ))
    temp.append(t)
max_temp=max(temp)
min_temp=min(temp)
max_day=days[temp.index(max_temp)]
min_day=days[temp.index(min_temp)]
print(f"Maximum temparature in week {max_temp} on {max_day}")
print(f"Minimum temparature in week {min_temp} on {min_day}")
```

```
IDLE Shell 3.14.6
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD6
] on win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\Lab4\temp.py ====
Enter the temperature of monday20
Enter the temperature of tuesday30
Enter the temperature of wednesday38
Enter the temperature of thursday27
Enter the temperature of friday34
Enter the temperature of saturday43
Enter the temperature of sunday24
Maximum temparature in week 43.0 on saturday
Minimum temparature in week 20.0 on monday
>>>
```

Result:The program is executed successfully ,taking input like temp and displayed max and min on which day.

Question 5: Write a Python program to remove duplicate elements from a list by preserving the order.

Program,Input&Output:

```
File  Edit  Format  Run  Options  Window  Help
n = int(input("Enter the number of elements: "))
lst = []
for i in range(n):
    item = input("Enter element: ")
    lst.append(item)

unique_list = []
for item in lst:
    if item not in unique_list:
        unique_list.append(item)
print("Original List:", lst)
print("List after removing duplicates:", unique_list)
```

```
IDLE Shell 3.14.6
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\Lab4\unique.py
Enter the number of elements: 4
Enter element: 1 1 2 3
Enter element: 1
Enter element: 1
Enter element: 2
Original List: ['1 1 2 3', '1', '1', '2']
List after removing duplicates: ['1 1 2 3', '1', '2']
>>>
```

Result:The program is executed successfully ,by removing duplicates from list.

Question-6: Create a program that manages a grocery shopping list

Program,Input&Output:

```python
grocery_list = ["Rice", "Milk", "Eggs", "Bread"]

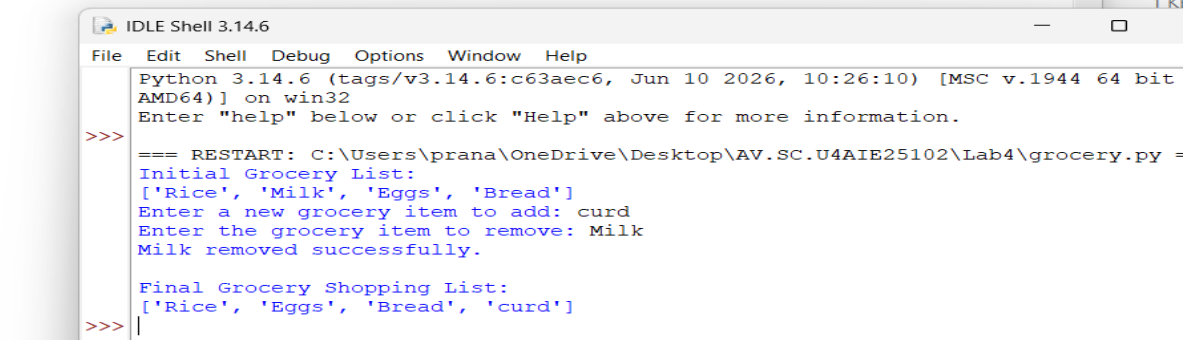
print("Initial Grocery List:")
print(grocery_list)

new_item = input("Enter a new grocery item to add: ")
grocery_list.append(new_item)

remove_item = input("Enter the grocery item to remove: ")

if remove_item in grocery_list:
    grocery_list.remove(remove_item)
    print(remove_item, "removed successfully.")
else:
    print(remove_item, "not found in the list.")

print("\nFinal Grocery Shopping List:")
print(grocery_list)
```



Result:The program is executed successfully ,by adding and removing list items.

Program-7: Write a menu-driven Python program to manage a list of books in a library

Program,Input&Output:

```
books = ["Python", "Java", "C Programming", "Data Structures"]

while True:
    print("\nLibrary Management System")
    print("1. Display all books")
    print("2. Add a new book")
    print("3. Insert a book at a specific position")
    print("4. Check availability of a book")
    print("5. Count occurrences of a book")
    print("6. Remove a book")
    print("7. Sort books alphabetically")
    print("8. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        print("\nBooks in Library:")
        for i, book in enumerate(books, start=1):
            print(i, ".", book)

    elif choice == 2:
        book = input("Enter the new book title: ")
        books.append(book)
        print(book, "added successfully.")

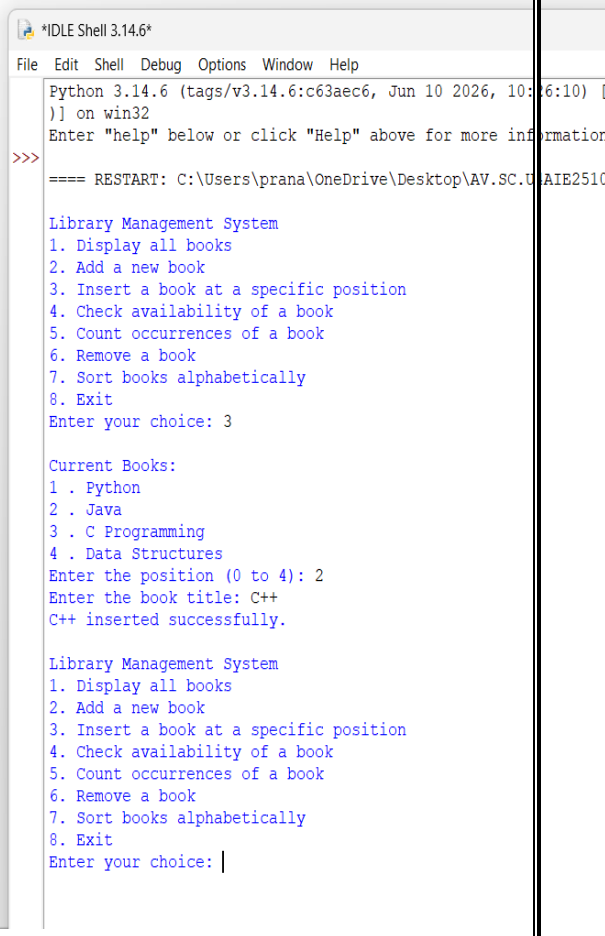
    elif choice == 3:
        print("\nCurrent Books:")
        for i, book in enumerate(books, start=1):
            print(i, ".", book)

        position = int(input("Enter the position (0 to " + str(len(books)) + "): "))
        book = input("Enter the book title: ")

        if 0 <= position <= len(books):
            books.insert(position, book)
            print(book, "inserted successfully.")
        else:
            print("Invalid position.")

    elif choice == 4:
        book = input("Enter the book title to check: ")

        if book in books:
            print(book, "is available at index", books.index(book))
        else:
            print(book, "is not available in the library.")
```



Result:The program is executed successfully in form of menu driven form.

Program-8:Write a Python program to demonstrate the use of list comprehension

Program,Input&Output:

```python
numbers = [10, 15, 20, 25, 30, 35, 40]

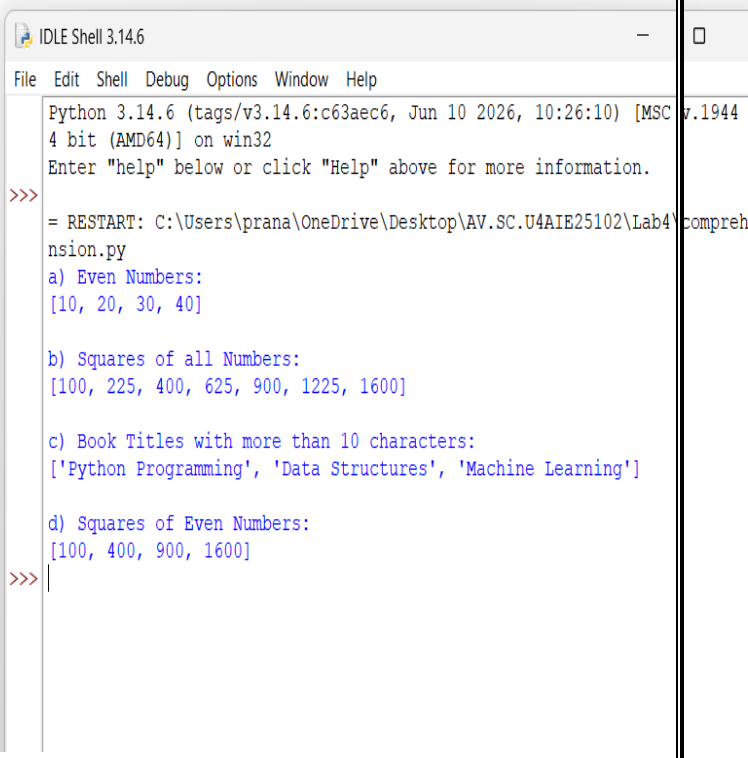
books = [
    "Python Programming",
    "Java",
    "Data Structures",
    "C",
    "Machine Learning"
]

even_numbers = [num for num in numbers if num % 2 == 0]
print("a) Even Numbers:")
print(even_numbers)

squares = [num ** 2 for num in numbers]
print("\nb) Squares of all Numbers:")
print(squares)

long_titles = [title for title in books if len(title) > 10]
print("\nc) Book Titles with more than 10 characters:")
print(long_titles)

even_squares = [num ** 2 for num in numbers if num % 2 == 0]
print("\nd) Squares of Even Numbers:")
print(even_squares)
```



Result:The program is executed successfully and known the list comprehension.

5. Overall Result

All 4 programs Executed Successfully used while and if elif in necessary conditions.